



PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2021
CASE NO. 3C2022

SUBJECT: BOTANY (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

Attempt FIVE Questions in All. Attempt in Urdu or English.

- Q. No. 1:** a) Explain Thallus organization and reproduction in Chlorophyta.
b) Give phylogeny /origin and classification of Ascomycetes.
(10+10=20 Marks)
- Q. No. 2:** a) Write note on Loose Smut of wheat and its control measures.
b) Give Auxospore formation and economic importance of Diatoms.
(10+10=20 Marks)
- Q. No. 3:** a) Mention general characteristics of Ginkgophyta.
b) Discuss comparison of structure of plant body of Lycopphyta with that of Psilophyta.
(10+10=20 Marks)
- Q. No. 4:** a) Write note on phylogeny of Cycads.
b) Explain Heterospory. Highlight its importance as a step towards seed habit.
(10+10=20 Marks)
- Q. No. 5:** a) Discuss Natural system of classification regarding its outline, merits and demerits.
b) Explain anatomy of bifacial and isobilateral leave, with reference to their dermal, ground and vascular systems.
(10+10=20 Marks)
- Q. No. 6:** a) Explain endosperm with reference to its types and nature.
b) Give structure, chemical composition and ultra-structure of plant cell wall.
(10+10=20 Marks)
- Q. No. 7:** a) Write detailed note on structure and function of sclerenchyma.
b) Differentiate between primary and secondary growth. Elaborate secondary growth in dicot stem with the help of suitable diagrams.
(10+10=20 Marks)
- Q. No. 8:** Write short notes on the following: (4x5=20 Marks)
i. Monocot stem and Dicot stem
ii. Monocot Root and Dicot Root
iii. Apomixies
iv. Polyembryony

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- Q. No. 6:**
- a) Describe the structure of plasma membrane in detail with diagrams.
 - b) Differentiate between followings
 - i) hypertonic and hypotonic solution
 - ii) Pinocytosis and phagocytosis
 - iii) exocytosis and endocytosis
 - iv) active transport and simple diffusion
 - v) primary and secondary lysosomes.
- (10+(5X 2)=20 Marks)**

SECTION-IV

- Q. No. 7:**
- a) What are different types of RNA's? Discuss their structure and role in the process of translation.
 - b) Define mutation. Write a detailed note on the kinds of point mutation and also discuss causes of mutation.
- (10+10=20 Marks)**

- Q. No. 8:**
- a) What is meant by sex-linkage? Discuss the pattern of inheritance of X linked traits with some suitable examples.
 - b) Write short notes on the following:
 - i) Barr bodies
 - ii) Post transcriptional changes in MRNA of eukaryotes:

(10+10=20 Marks)

SECTION-V

- Q. No. 9:**
- a) What is Organic Evolution? How did the following contribute to Charles Darwin's formulation of the theory of natural selection? i) Uniformitarianism ii) Malthus essay on "Principles of population"
 - b) Write note on the followings:
 - i) Neo-Darwinism
 - ii) Adaptive radiation

(10+(5x2)=20 Marks)

- Q. No. 10:**
- a) What is the theory of inheritance of acquired characteristics and how did Lamarck use it to explain the phenomenon of evolution?
 - b) What are the four assumptions of Hardy- weinberg law? How do the allelic frequency of population is changed? Explain with some suitable examples.

(10+10=20 Marks)
